

EE / CprE / SE 491 — sdmay27-11

Weekly Report 2:

MIMO FMCW Radar

28th September – 3rd October 2026

Client and Faculty Advisor: Dr. Mohammed Tayeb Al Qaseer

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§1 Summary

This week required us to take a look at the work we were doing and manage some of the risks we were taking with the DACs which were way overkill for our application currently and just trying to find a minimum viable product that we can use well. Because of that we designed a new simplified DAC with the Vivaldi being evaluated as the microstrip-to-waveguide transition; it is not yet matched across 24 to 40 GHz. Signal processing defined the IF capture chain, and the FPGA sub-team set up an Opal Kelly board with blinky and planned the SPI controller.

§1.1 Team and Roles

Team Member	Role
Flatrim Dreshaj	Team Leader
Justin Adrowski	FPGA Programming
Durotimi Johnson	TX/RX Layout Design
Ryland Feist	VCO / PLL / Synthesizer
Elliott Bourgeois	Signal Processing
Brett Brunner	Antenna Design
Aidan O'Grady	FPGA / MCU Programming

Table 1. Team members and roles

§1.2 Status at a Glance

Work Item	Owner	Status
Simplified DAC board	Flatrim Dreshaj	Schematic and layout complete
DAC fast-sweep board	Flatrim Dreshaj	Schematic and layout complete
Ethernet SPI board	Flatrim Dreshaj	In Review(possibly unnecessary)
VCO board	Ryland Feist	Schematic and layout complete
TX up-converter board	Durotimi Johnson	Schematic complete, layout in progress
Waveguide and transition design	Flatrim Dreshaj	In progress
Antenna Design	Brett Brunner	In progress
IF capture chain	Elliott Bourgeois	Architecture defined; algorithms in progress

Work Item	Owner	Status
FPGA SPI controller	Justin Adrowski, Aidan O'Grady	Research done, not yet programming

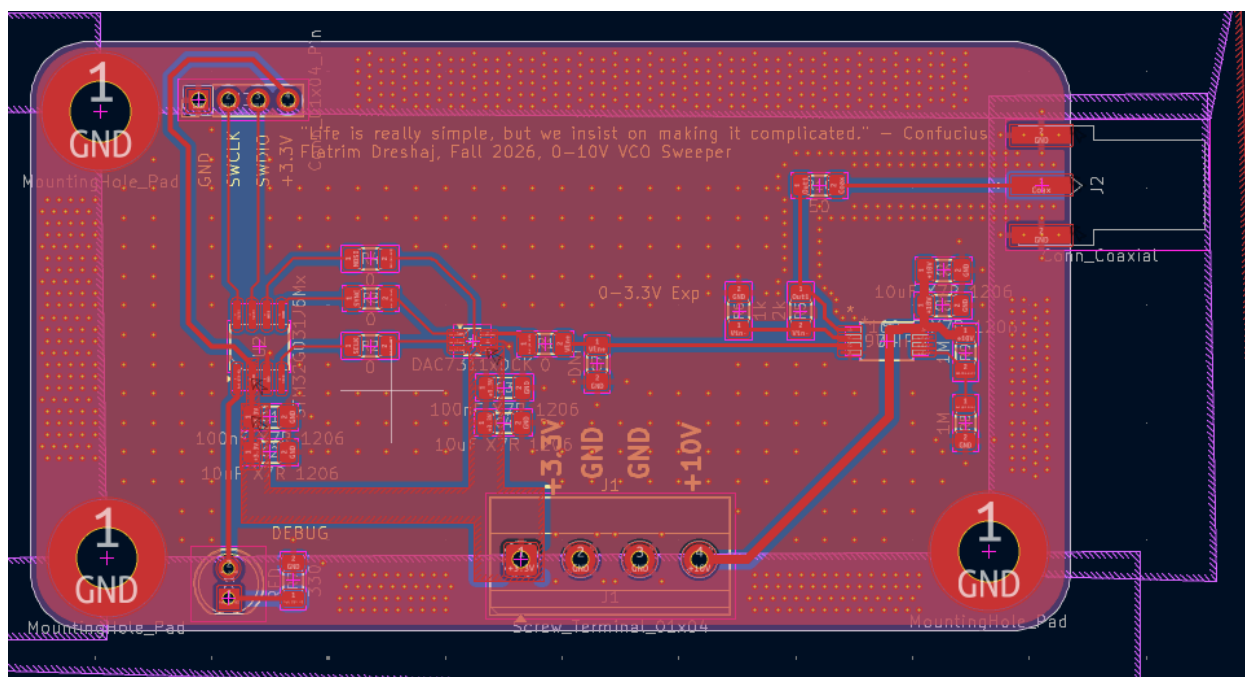
Table 2. Status of each work item at the end of the week

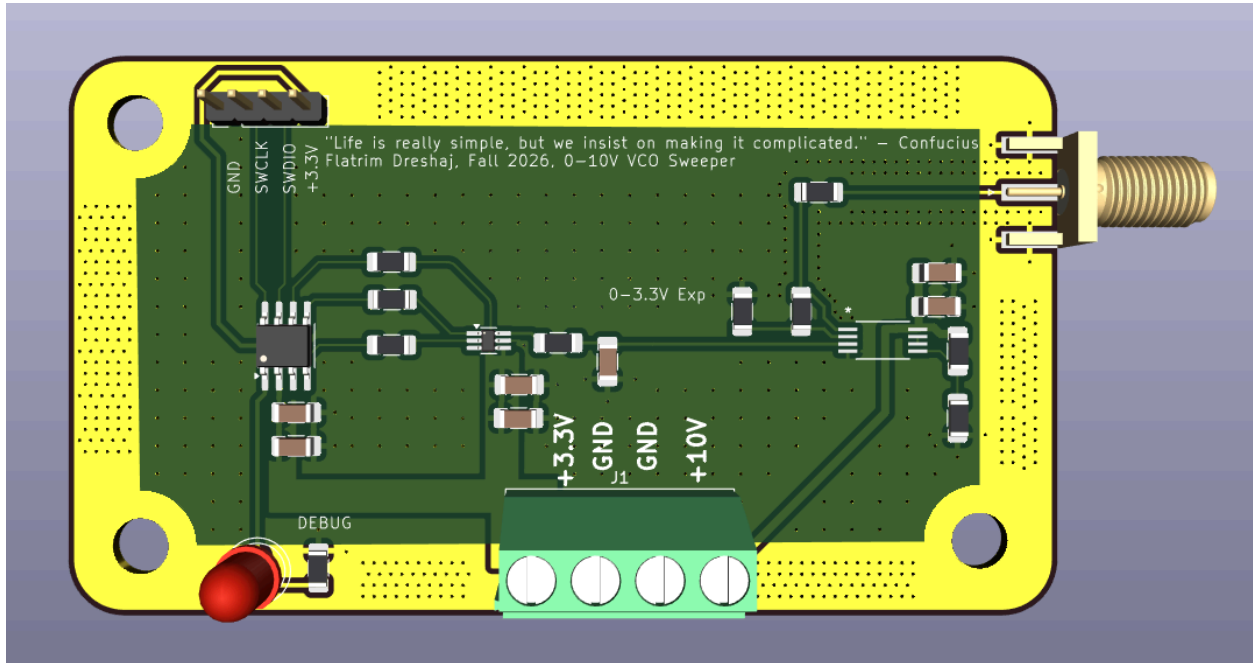
§2 Feedback from last report

We were given feedback to provide the background of the project and at what stage in the SD Team picking it up. As it relates to this project we had the idea for it during the summer while talking with Dr Tayeb, since we had thought of it during that time the us together with Dr Tayeb already had a vision in mind as to what we needed to get done and what each subsystem would look like. Given that we started early as soon as the semester started and we got enough people.

§3 PCB Design

Big goal for us is trying to minimize the risk with building these board so we are going through and making some simpler boards that require much less to get off the ground running. The DAC is an example of this where instead of the previous design which used a current output DAC, we are now using a voltage output DAC swept through SPI by the smallest STM32 MCU I could find which we program through SWD. The output of the DAC is 0-3.3V so I just amplify that through a non inverting amp with Gain of 3 which gives us 0-10V. GBP is not an issue here since our sweep times for this initial test will be max 1/20 Hz which is trivial.





§3.3 VCO Board

The VCO board (Figure 8) has screw terminals for power and tuning, the HMC587LC4B, decoupling capacitors, and a resistive splitter feeding three SMA outputs.

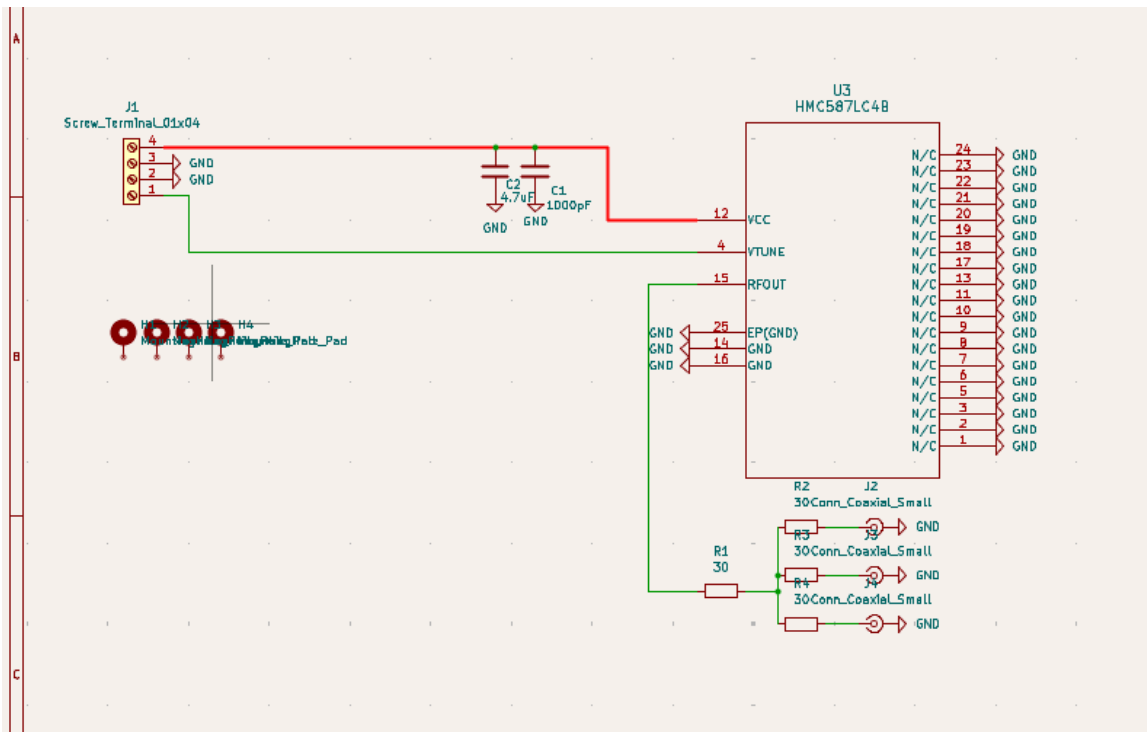


Figure 8. VCO board schematic with the HMC587LC4B and a three-way resistive splitter

The splitter resistors follow the matched star-splitter formula for N ports:

$$R = Z_0 (N - 1) / (N + 1) \quad (2)$$

With $N = 4$ and $Z_0 = 50 \, \Omega$ this gives the $30 \, \Omega$ used. Each output is about 9.5 dB below the input, which sets the drive level at the up-converter.

In the layout (Figure 9), the $50 \, \Omega$ lines to the SMA connectors are via-fenced and have the solder mask removed.

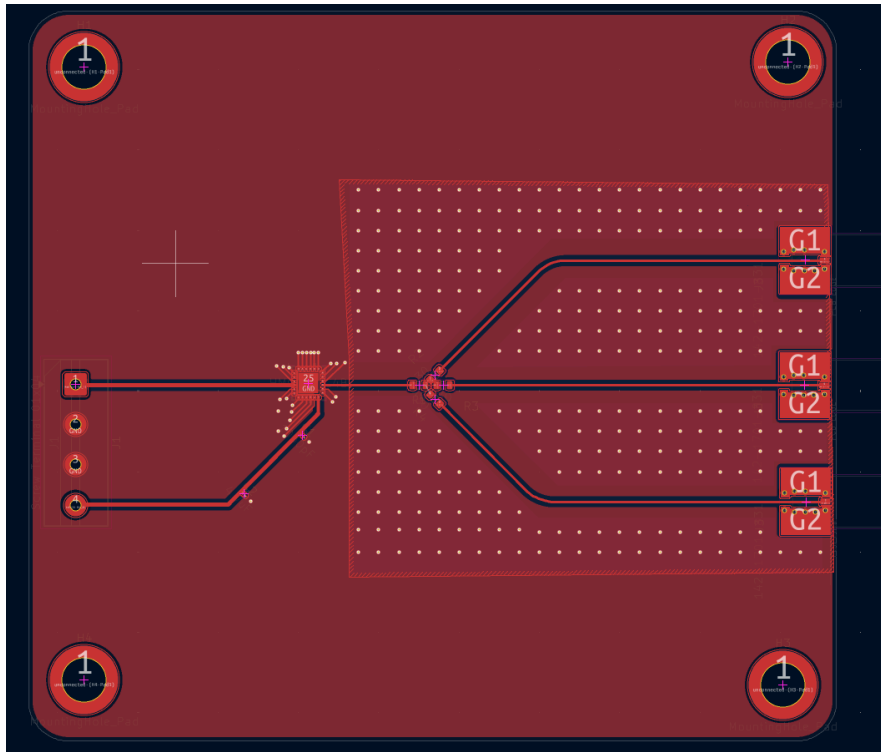
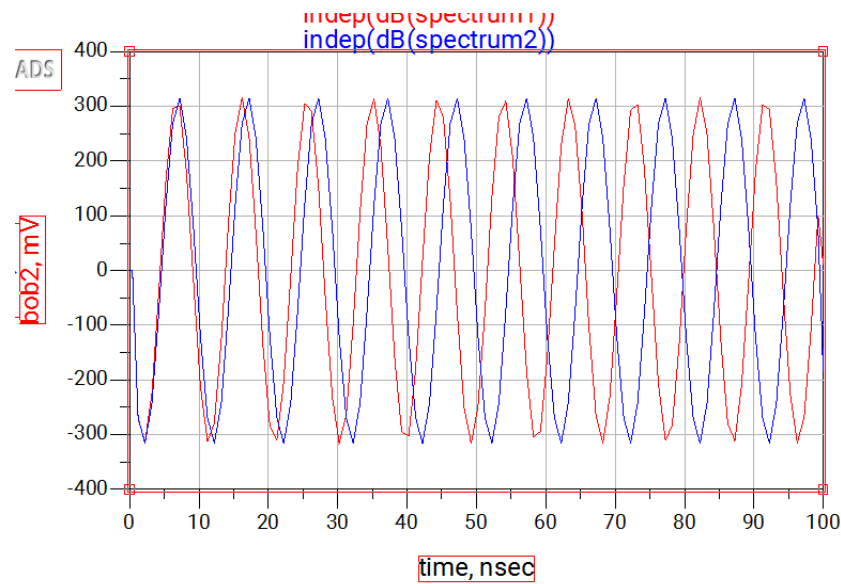


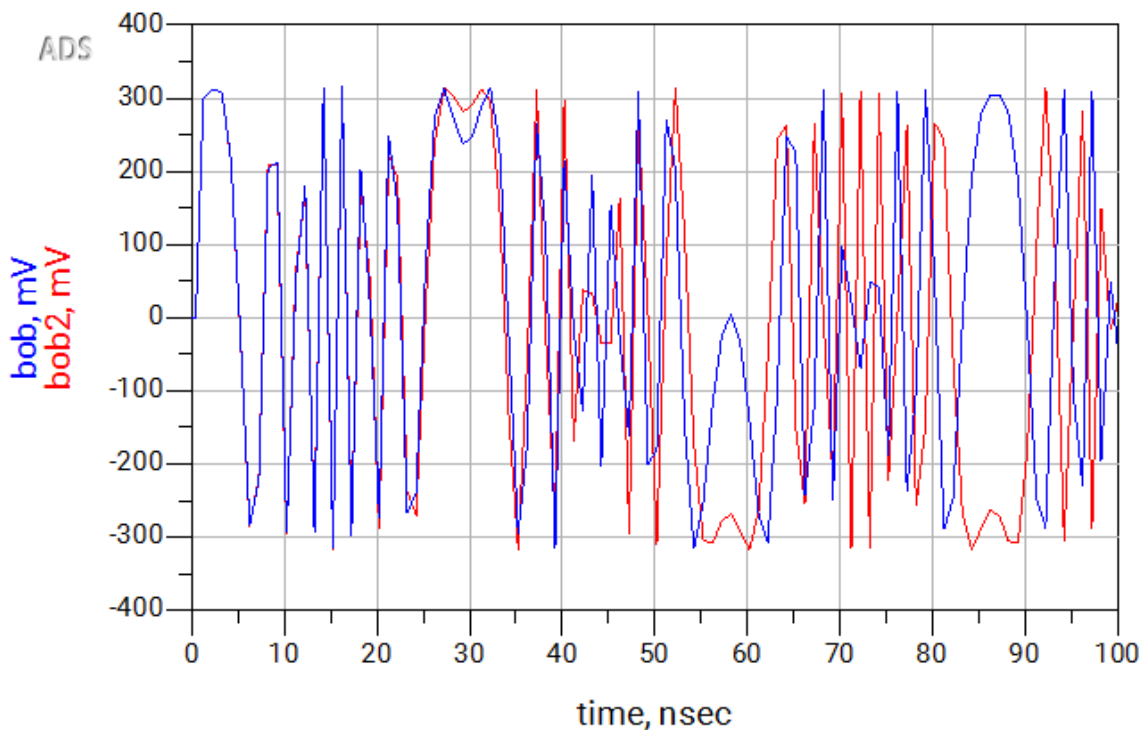
Figure 9. VCO board layout with via-fenced $50 \, \Omega$ lines to the SMA outputs

§3.31 PLL Research

The DAC and VCO setup is going to have some frequency drift due to the increase in temperature which will inevitably follow long operating times. The VCO also experiences some change in voltage sensitivity(MHz/V) for its vtune input as temperature increases. Because of this, we first determined whether it was going to be an issue that would cause us problems, and using some ADS simulations with ideal VCOs As it turns out at any one frequency the drift does not seem to matter that much, asides from some phase issues.



The problem is that when you start to sweep the range the difference in the frequency compounds, which is where we would develop issues, should we calibrate the FPGA for when the device is cold.



The solution is to either not operate the radar when it is not at operating temperature, or to use a PLL/synthesizer to prevent frequency drift. The PLL Synthesizer we are currently looking at is the ADF41510, as it can operate up to 10GHz, and it would give us a minimum

output resolution of around 300Hz. We will also be looking at a solution where we intentionally heat the VCO to a specific temperature such that it is less sensitive to minor heat fluctuations.

§3.4 TX Up-Converter Board

We decided to change a lot of various things. We decided to switch from horizontal launch SMA connectors to vertical launch 2.92mm connectors and the positive outs for the outputs are going to be connected with a 50 ohm termination since we're just going to be using the negative outs instead to keep the lines straight. We'll be losing about half the signal because of that, but we think it's worth it.

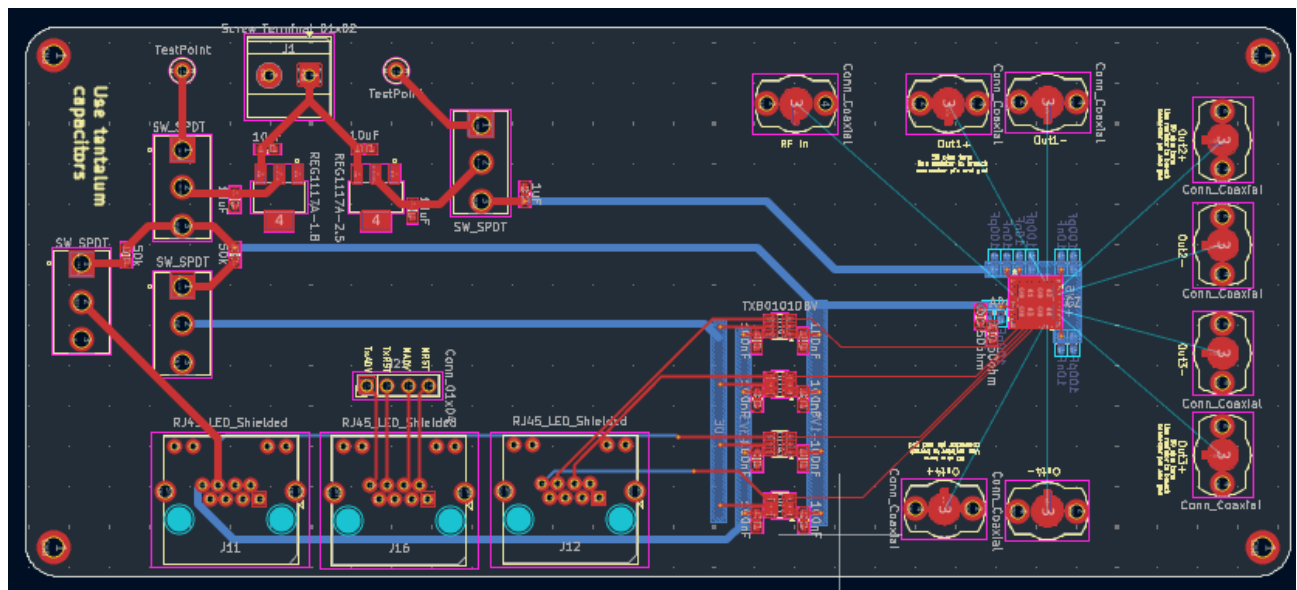


Figure X: Updated Layout w/ visual of 2.92mm Connectors

We put the components that are close to the ADAR2001 to the bottom side of the PCB to free up a lot of space close to the ADAR. Since we're doing a 4 layer board, the grounding under the RF lines won't be an issue.

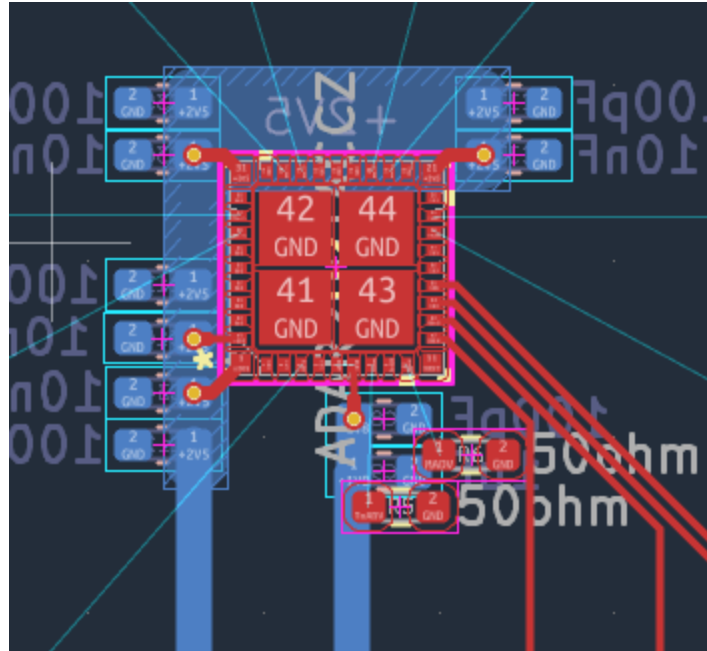


Figure X: Close visual of ADAR section

Since the level shifters are going to be very close to each other and a lot of their pins will be connected to the same node, we decided to use copper pours on the bottom layer of the board.

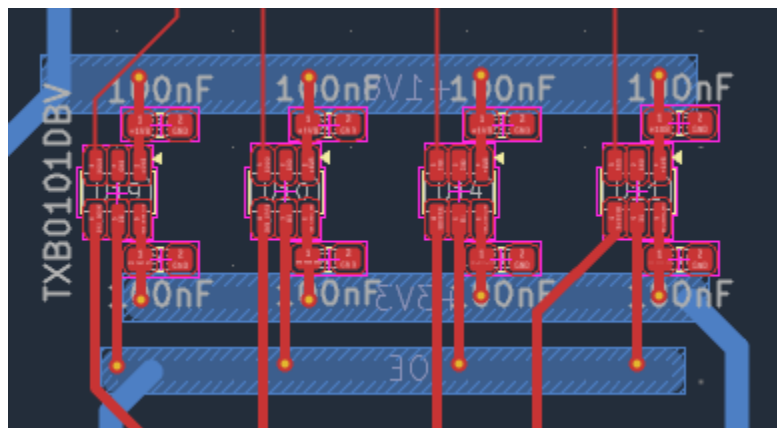


Figure X: Copper Pours For Level Shifters

§4 Waveguide and Antenna Design

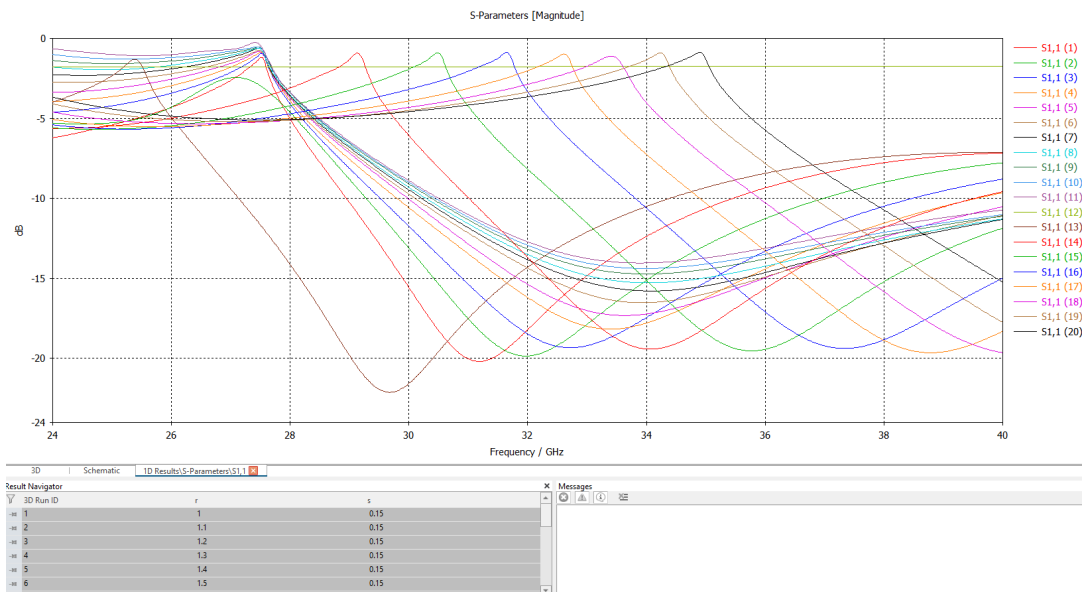
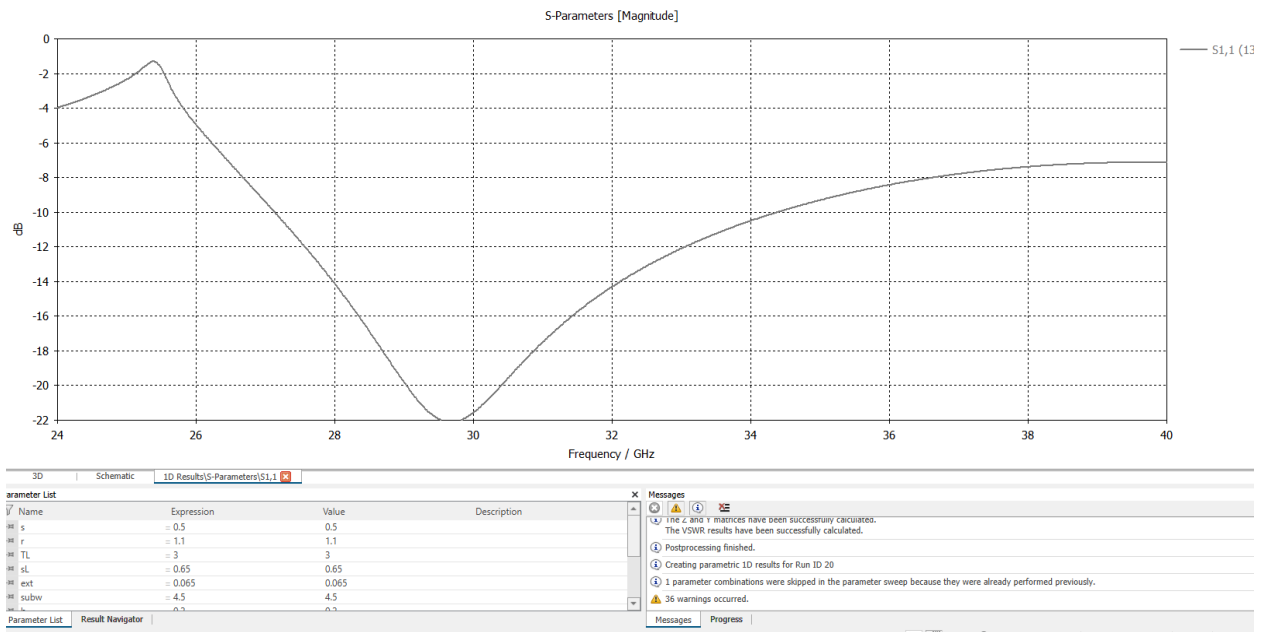
The vivaldi antenna needs a microstrip to slotline transition, and various variables have been adjusted to accommodate the S11 parameter.

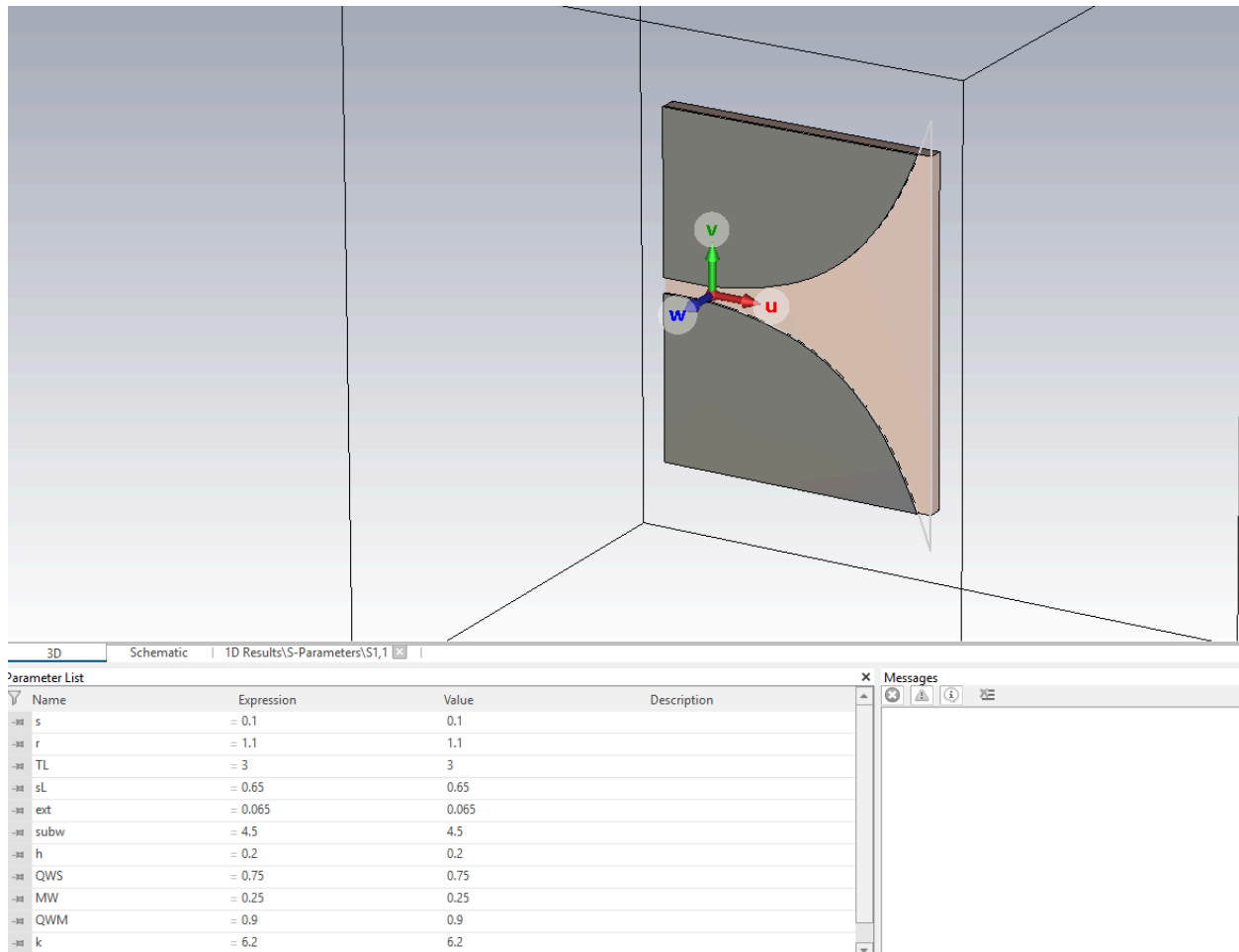
§4.1 Waveguides and Transitions

Work this week covered tuning of the vivaldi, specifically the opening curvature taper, radius of the slotline, overall board length and distance before the exponential flare.

§4.2 Vivaldi as a Waveguide Transition

The goal continues to be S11 below -10 dB from 24 to 40 GHz. Instead of running the optimizer, the antenna was tuned via sweeping a parameter, choosing its best S11 result and then adjusting other variables with the new choice as a constant.





\$5 Signal Processing

Until the FPGA ADC path is running, the lab spectrum analyzer will sample the IF signal from the receiver board and pass it to a PC (Figure 20). This lets algorithm development start now.

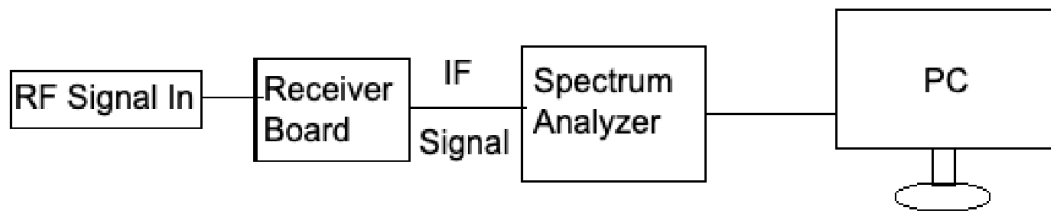


Figure 20. Planned IF capture chain using the lab spectrum analyzer

This week, I spent time setting up the interface between the spectrum analyzer and the PC via the API. The API is created for C-language interfacing, but we will be using Python. Using Python's ctypes and an .py file, we are able to use Python to interact with the spectrum analyzer via this API. Thus far, I have been able to collect samples from the spectrum analyzer and compute FFTs on those samples.

This process involves communicating with the spectrum analyzer, setting the center frequency and bandwidth, as well as the sample rate/communication rate, then sampling for a fixed number of samples. All of these are configurable via Python, allowing automation of data collection to test DSP algorithms before the rest of the system is functioning.

Moving forward, I will be exploring our back-up method of data acquisition, using an NI DAQ card, which will enable data acquisition via integrated ADCs. This will serve as an alternative to the microcontroller method we are currently exploring as well.

§6 FPGA Development

The sub-team researched SPI controller architectures for a VHDL implementation, which will be developed on an Opal Kelly board with ADC modules (Figure 21).

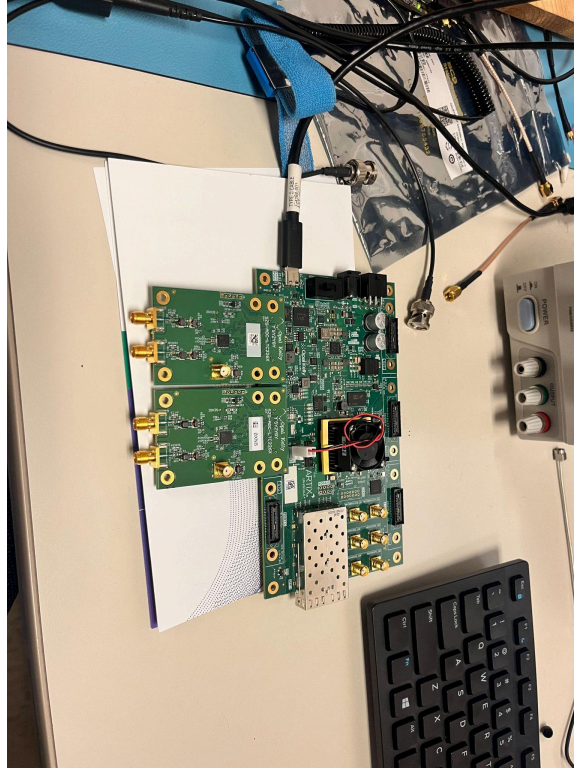


Figure 21. Opal Kelly development board with ADC modules attached

We have begun interfacing with the board, getting a basic blinky program to run. The next step is to map pins on a port-breakout board to an out-of-the-box spi master VHDL implementation we found online. We can then use the SPI protocol to interface with an STM32 configured as a SPI slave to print to putty or some console.

§7 Research

- **Datasheets:** AD3551R layout guidelines, TXB0101 and REG1117, applied in the schematics above.
- **SPI controller architectures:** reviewed for the VHDL design.
- **PLL research:** for the VCO and synthesizer.
- **Waveguides and transitions:** waveguide cutoff and microstrip-to-waveguide transitions, including tapered-slot and finline types.
- **Vivaldi design:** how the flare geometry sets the lowest usable frequency.

No measurements were taken this week. Primary data collection starts once spectrum analyzer sampling is set up and the first boards are assembled.

§8 Pending Issues

- Board Fab and Assembly
- FPGA vs STM32H7
- **Waveguide transition:** the Vivaldi is a candidate, not yet final, and must match well at 24 GHz.
- **Vivaldi match:** S11 is not yet below -10 dB across 24 to 40 GHz (best about -6.4 dB, worst about -2 dB).
- **ADAR placement:** still open. It sets the RF trace lengths for the eight equal-length TX outputs.
- **FPGA programming path:** Opal Kelly SDK or JTAG through Vivado.
- **TX up-converter:** the schematic may change, and the layout is not finished.

§9 Individual Contributions

Team Member	Contribution	Weekly Hours	Total Hours
Flatrim Dreshaj	Simplification of systems and adding redundancies, systems design, team organization	5	50
Brett Brunner	Antenna design	7	16
Durotimi Johnson	TX up-converter board	6	26
Ryland Feist	VCO board, PLL research	3.5	13.5
Elliott Bourgeois	Signal processing	6	14
Justin Adrowski	FPGA	3	7
Aidan O'Grady	FPGA	4	12
Team Total			

Table 4. Individual contributions and hours

§10 Plans for Next Week

- **Do CST simulations of differential lines, connectors, terminations for the ADAR board. (Flatrim)**
- **Antenna (Brett):** get S11 below -10 dB across 24 to 40 GHz by widening the flare and running the optimizer. Study antenna concepts and research other candidates for end goal antenna.

- **Signal processing (Elliott):** Start DSP algorithm development and explore the NI DAQ cards.
- **TX up-converter (Durotimi):** continue the layout with equal-length RF routing.
- **FPGA (Justin, Aidan):** settle the programming path and start the VHDL SPI controller. Look into alternatives to FPGA, such as STM32H7/F7 series chips.
- **VCO (Ryland):** PLL board schematic and research.